

VOLCHIKHA, Russian Federation

WMO: 360220

Lat: 52.022N

Lon: 80.340E

Elev: 208

StdP: 98.85

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.1	-34.2	-41.0	0.1	-38.0	-37.0	0.1	-34.0	11.0	-6.9	9.7	-6.4	1.4	40	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	31.8	18.4	30.0	17.8	28.2	17.2	20.3	27.4	19.4	26.7	18.6	25.6	4.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.4	22.7	17.0	12.5	21.6	16.1	11.7	21.3	59.5	27.5	56.4	26.5	53.7	25.9	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	7.8	DB	-40.4	35.4	4.0	1.8	-43.3	36.6	-45.6	37.7	-47.8	38.7	-50.7	40.0	(4)
(5)			WB	-40.4	21.9	3.8	1.9	-43.2	23.3	-45.4	24.5	-47.6	25.6	-50.4	27.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-17.6	-15.0	-6.6	5.4	12.7	18.6	20.1	17.8	11.4	4.2	-5.7	-13.3	(6)	
	DBStd	14.66	9.11	8.19	7.18	5.93	4.98	3.88	3.29	3.73	4.57	5.09	8.17	8.24	(7)	
	HDD10.0	3678	854	700	513	158	31	1	0	1	37	190	471	723	(8)	
	HDD18.3	5863	1113	933	772	388	185	44	20	55	212	439	721	982	(9)	
	CDD10.0	1033	0	0	0	21	113	258	312	242	79	8	0	0	(10)	
	CDD18.3	176	0	0	0	1	9	51	73	38	4	0	0	0	(11)	
	CDH23.3	2295	0	0	0	18	197	641	758	564	115	1	0	0	(12)	
	CDH26.7	738	0	0	0	3	53	205	250	194	33	0	0	0	(13)	
(14) Wind	WSAvg	3.6	3.4	3.4	3.8	4.1	4.1	3.5	3.2	3.2	3.3	3.5	3.8	3.7	(14)	
(15) Precipitation	PrecAvg	351	18	15	21	21	26	40	62	34	26	30	32	26	(15)	
	PrecMax	473	35	33	72	56	92	121	144	126	77	84	63	47	(16)	
	PrecMin	190	6	1	4	0	9	7	14	5	3	10	12	11	(17)	
	PrecStd	74	8	8	14	15	17	25	33	24	16	17	13	10	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	2.2	12.0	25.9	31.5	33.2	34.1	34.6	30.5	22.1	12.2	2.2	(19)	
		MCWB	0.1	0.2	5.9	12.7	16.4	18.0	19.6	18.8	16.1	11.5	6.9	0.6	(20)	
	2%	DB	-0.5	0.3	7.2	21.7	28.0	30.9	31.5	31.2	26.7	18.2	8.6	0.5	(21)	
		MCWB	-1.8	-1.3	3.6	11.3	14.8	17.7	19.3	18.1	14.6	9.9	5.1	-0.7	(22)	
	5%	DB	-2.8	-1.3	4.5	18.6	25.6	29.0	29.4	28.7	23.5	15.3	5.8	-1.0	(23)	
		MCWB	-4.0	-2.8	1.9	9.8	13.9	17.3	18.6	17.6	13.3	8.7	3.2	-2.1	(24)	
	10%	DB	-4.9	-3.2	2.6	15.6	23.0	27.1	27.5	26.6	20.5	12.2	3.5	-2.5	(25)	
		MCWB	-6.0	-4.5	0.7	8.4	13.0	17.0	18.3	16.7	12.4	7.2	1.7	-3.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	0.7	6.7	14.0	17.6	20.3	22.0	20.7	17.1	12.4	8.0	1.0	(27)	
		MCDB	1.5	1.6	10.8	23.4	27.4	27.9	29.1	29.4	27.5	19.3	10.8	1.9	(28)	
	2%	WB	-1.6	-1.2	3.9	12.2	16.1	19.2	20.9	19.3	15.3	10.7	5.2	-0.5	(29)	
		MCDB	-0.6	0.0	6.7	19.7	25.2	27.0	27.2	28.2	24.0	16.8	8.1	0.3	(30)	
	5%	WB	-3.9	-2.7	2.2	10.5	14.9	18.5	20.0	18.4	14.2	9.2	3.4	-2.0	(31)	
		MCDB	-2.9	-1.4	4.2	17.2	23.3	26.2	26.5	26.4	22.3	14.6	5.8	-1.1	(32)	
	10%	WB	-6.0	-4.5	0.8	9.0	13.7	17.7	19.2	17.5	12.9	7.7	1.7	-3.5	(33)	
		MCDB	-5.0	-3.2	2.4	15.1	21.4	25.1	25.7	24.8	19.6	11.7	3.5	-2.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.6	11.0	12.5	13.8	12.7	11.8	13.5	13.3	10.6	8.2	9.4	(35)	
		MCDBR	9.3	9.5	11.0	17.6	18.4	16.4	15.2	17.6	19.2	17.0	9.6	8.5	(36)	
	5% WB	MCWBR	8.7	8.3	7.9	8.9	8.1	6.4	5.5	6.7	8.7	9.7	6.6	8.1	(37)	
		MCWBR	9.5	9.1	9.7	15.9	15.2	13.5	12.4	15.0	16.5	15.3	8.7	8.5	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.336	0.388	0.410	0.415	0.413	0.436	0.414	0.373	0.354	0.339	0.307	(40)	
		taud	2.031	1.954	1.926	2.143	2.184	2.275	2.212	2.243	2.303	2.254	2.098	1.990	(41)	
	Ebn at Noon	Ebn at Noon	656	738	776	821	837	842	813	810	802	730	600	557	(42)	
		Edh at Noon	81	118	149	136	137	127	133	122	102	88	77	72	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.27	2.38	3.84	4.73	5.99	6.39	5.79	5.09	3.73	2.02	1.04	0.87	(44)	
		RadStd	0.10	0.12	0.32	0.35	0.43	0.46	0.39	0.30	0.31	0.27	0.12	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.46	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page